Jiacheng Tian

PhD student in Geophysical Fluid Dynamics Group at ETH Zurich
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Education

PhD in Geophysics, ETH Zurich, Switzerland — 11.2021 – Present
- Thesis: Linking geochemical observations to Earth's mantle dynamics: insights from geodynamic models
- Advisor: Prof. Paul Tackley

MSc in Earth Sciences (with distinction), ETH Zurich, Switzerland — 09.2018 – 09.2021
- Thesis: The influence of a weak crust on Venus tectonics
- Advisor: Prof. Paul Tackley

BSc in Geology, Peking University, China — 09.2013 – 06.2017

Selected Publications

2. **Tian, J.**, Lourenço, D. L., & Tackley, P., (2025). Influence of lower mantle thermal conductivity on the thermal evolution of Earth's mantle and core. *Earth and Planetary Science Letters*, *672*, 119695.
1. **Tian, J.**, Tackley, P., & Lourenço, D. L. (2023). The tectonics and volcanism of Venus: New modes facilitated by realistic crustal rheology and intrusive magmatism. *Icarus*, *399*, 115539.

Awards

Chun-Tsung Scholar, Chun-Tsung Endowment	2017
May 4th Scholarship, Peking University	2015
Robin Li Scholarship, Peking University	2014
Silver Medal, 27th Chinese Chemistry Olympiad (CChO)	2013

Teaching

Gravimetry (TA, BSc field course)	2023 – 2025
Introduction to finite element modelling in geosciences (TA, MSc course)	2023

Outreach

2025 German-Swiss Geodynamics Workshop	Organizer (ETH GFD Group)
2022 Ada Lovelace Workshop	Co-organizer (ETH GFD Group)

Selected Conference Presentations

Oral:

6. **Tian, J.**, Tackley, P. Effects of thermal conductivity on the long-term thermal evolution of the lower mantle and the core, *AGU 2023*
5. **Tian, J.**, Tackley, P., & Rozel, A. Implications of a realistic crustal rheology and intrusive magmatism on Venusian tectonics, *EGU 2022*

Poster:

4. **Tian, J.**, Tackley, P. & Elliott, T. Linking the evolution of short-lived isotopic ratios in mantle to early Earth dynamics: Insights from geodynamic models, *Goldschmidt 2025*
3. **Tian, J.**, Tackley, P. & Elliott, T. Tracking the evolution of isotopic heterogeneities in early Earth with mantle convection models, *AGU 2024*
2. **Tian, J.**, & Tackley, P. Long-term preservation of geochemical heterogeneities in early Earth: tracking short-lived isotopes in geodynamic models, *EGU 2023*.
1. **Tian, J.**, & Tackley, P. The evolution of isotope heterogeneities in early Earth: A geodynamic perspective, *AGU 2022*.

Invited Talk

Earth and planetary science seminar, *University of Chinese Academy of Sciences* — 11.2025

Skills

Programming
- **Programming languages**: MATLAB, Fortran, Julia, Python
- **Tools**: Git, Paraview, Bash script

Language: Chinese (native); English (fluent)